

RYAN HEEJOON BYUN

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EDUCATION

Seoul National University

M.S. in Statistics

Overall GPA: 4.25/4.3 (through Spring 2026)

Thesis advisor: Won Chang

Thesis: Miscoverage Risk Controlled Computer Model Calibration

March 2025 – February 2027 (Expected)

Seoul National University

B.S. in Electrical & Computer Engineering

Overall GPA: 4.12/4.3

Summa Cum Laude

March 2020 – February 2024

RESEARCH INTERESTS

Valid black-box-assisted inference; conformal and distribution-free inference; calibrated Bayesian and simulation-based inference; uncertainty quantification for scientific simulators and neural surrogates.

PUBLICATIONS AND MANUSCRIPTS

Manuscripts

- [1] **Heejoon Byun**, Dayeon Yoon, Won Chang. “Miscoverage Risk Controlled Computer Model Calibration.” Manuscript in preparation; preprint expected Fall 2026.
Develops a conformal risk control framework for calibrated posterior predictive uncertainty in amortized computer model calibration under data–model discrepancy.

Conference Proceedings

- [2] Hwihun Jeong, **Heejoon Byun**, Dong Un Kang, Jongho Lee. “BlindHarmony: Blind Harmonization for MR Images via Flow Model.”
In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

RESEARCH EXPERIENCE

Statistical Inference and Scientific Model Calibration

Master’s Researcher; Advisor: Won Chang

March 2025 – Present

Seoul National University

- Developed an amortized computer model calibration framework combining conditional flow matching with conformal risk control for calibrated posterior predictive uncertainty.
- Designed simulation studies and a real-data SEIR calibration analysis to evaluate predictive coverage under data–model discrepancy.
- Implemented neural posterior approximation, conformal calibration, and MCMC baseline comparisons in NumPy and PyTorch.

Biophotonics and Nano Engineering Lab (BiNEL)

Research Intern; Advisor: Sunghoon Kwon

July 2023 – December 2023

Seoul National University (SNU), Korea

- Developed a protein-language-model-based classifier for immune repertoire data and implemented model training and evaluation pipelines in PyTorch.

Laboratory for the Imaging Science and Technology (LIST)

Research Intern; Advisor: Jongho Lee

December 2022 – March 2023

Seoul National University (SNU), Korea

- Developed a neural-spline-normalizing-flow-based MR image harmonization algorithm; work resulted in an ICCV 2023 publication.

TEACHING EXPERIENCE

Statistical Experiments with R

Teaching Assistant

Fall, 2025

Seoul National University

- Lectured recitation sessions; designed homework and exam problems; graded assignments and exams; assigned final grades.

Introduction to Data Science

Teaching Assistant

Spring, 2025

Seoul National University

- Designed homework problems and graded assignments.

SELECTED COURSEWORK

Statistics and Probability: Graduate Statistical Theory I–II (A+, A+); Graduate Probability Theory I–II (A+, A+); Applied Statistics (A+, ranked 1st); Advanced Statistical Computing (Ao).

Mathematics: Functional Analysis (A+); Differentiable Manifolds (A+); Introduction to Mathematical Analysis (A+); Linear Algebra (A+).

Machine Learning and Computation: Mathematical Foundations of Deep Neural Networks (A+); Reinforcement Learning (A+); Information Theory (A+); Computer Vision (A+); Data Structures (A+); Digital Signal Processing (A+); Signals and Systems (A+).

Selected distinctions: Only A+ in Graduate Probability Theory I; ranked 1st in Applied Statistics.

HONORS & AWARDS

National Scholarship for Science and Engineering, KOSAF

Merit-based national scholarship for science and engineering students.

June 2022 – August 2023

Merit-Based Scholarship, Seoul National University

Awarded for academic performance.

August 2020 – August 2021

TECHNICAL SKILLS

Programming

Python, R, MATLAB, C++

Statistical/ML Computing

PyTorch, NumPy, scikit-learn